
Mathematical and Computational Methods

Exercise Sheet 4: Vectors

Tutorial route (▶, in this order, about 75 minutes): 1.3(b,c), 2.2, 2.4, 2.7, 2.8(a,b), 2.9, 2.10. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 (*★ drill — component arithmetic and a sketch*). Let $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 2)$ in $\mathbb{R}^2$.

- (a) Compute $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $3\mathbf{u}$ and $2\mathbf{u} - 3\mathbf{v}$.
- (b) On one set of axes sketch $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$, each drawn from the origin, and check your first two answers to (a) against the sketch.

Exercise 1.2 (*★ drill — vector between points, distance*). For the points $A(2, -1, 3)$ and $B(5, 1, -1)$, find (a) the vector $\overrightarrow{AB}$ and (b) the distance from A to B .

Exercise 1.3 (*► ★ drill — unit vectors*). (a) Normalise $\mathbf{v} = (3, 4)$.

- (b) Normalise $\mathbf{w} = (1, -2, 2)$.
- (c) Find all unit vectors in $\mathbb{R}^2$ that make an angle of 45° with $(1, 0)$.

Exercise 1.4 (*★ drill — dot product, orthogonality and angle*). Let $\mathbf{a} = (2, 3)$, $\mathbf{b} = (3, -2)$, $\mathbf{c} = (1, 1)$ and $\mathbf{d} = (-1, 0)$.

- (a) Compute $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$, and say which pair (if any) is orthogonal.
- (b) Find the angle between $\mathbf{c}$ and $\mathbf{d}$.

Exercise 1.5 (*★ drill — projection*). Project $\mathbf{u} = (6, 2)$ (a) onto $\mathbf{v} = (3, 0)$ and (b) onto $\mathbf{w} = (0, 5)$.

Exercise 1.6 (*★ drill — cross product and area*). (a) Compute $\mathbf{i} \times \mathbf{j}$.

- (b) For $\mathbf{u} = (2, -2, 1)$ and $\mathbf{v} = (1, 2, 2)$ compute $\mathbf{u} \times \mathbf{v}$ and check that it is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.
- (c) Hence find the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$, and of the triangle with these as two sides.

Exercise 1.7 (*★ drill — equation of a line*). (a) Give the vector, parametric and symmetric equations of the line through $P(1, 2, 3)$ in the direction $\mathbf{d} = (2, -1, 4)$.

- (b) Give the vector equation and the slope of the line through $A(0, 1)$ and $B(4, 3)$ in $\mathbb{R}^2$.

Exercise 1.8 (*★ drill — equation of a plane*). (a) Find the Cartesian equation of the plane with normal $\mathbf{n} = (1, 2, -1)$ through the point $(2, -1, 3)$.

- (b) Write down a normal vector to the plane $3x - y + 2z = 7$.

2 Tutorial problems

Exercise 2.1 (★★core — linear combinations). Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.

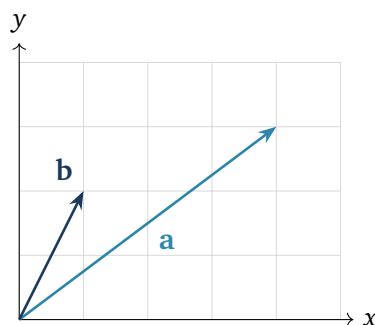
- Express $\mathbf{c} = (7, 0)$ as $\alpha\mathbf{a} + \beta\mathbf{b}$.
- Are $\mathbf{a}$ and $\mathbf{b}$ linearly independent? Explain geometrically.

Exercise 2.2 (►★★core, data science — angle and cosine similarity). Over the four-word vocabulary (data, model, train, plot) two short documents have word-count vectors $\mathbf{u} = (1, 1, 0, 0)$ and $\mathbf{v} = (0, 1, 1, 0)$.

- Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$ using the dot product.
- State their cosine similarity $\cos \theta$ and interpret it: on the scale from 0 (no words in common) to 1 (the same word proportions), how similar are the documents?

Exercise 2.3 (★★core, conceptual — spot the error). A student writes: “For $\mathbf{u} = (1, 2, 2)$ and $\mathbf{v} = (2, 2, 1)$ we have $\mathbf{u} \cdot \mathbf{v} = 2 + 4 + 2 = 8$, so $\cos \theta = 8$ and $\theta = \arccos 8$.” Identify the mistake and give the correct angle.

Exercise 2.4 (►★★core — reading a diagram: parallel and perpendicular parts). The figure shows two vectors $\mathbf{a}$ and $\mathbf{b}$ drawn from the origin on a unit grid.



- Read off the components of $\mathbf{a}$ and $\mathbf{b}$ from the grid.
- Resolve $\mathbf{a}$ into a component $\mathbf{a}_{\parallel}$ parallel to $\mathbf{b}$ and a component $\mathbf{a}_{\perp}$ perpendicular to it. Verify that the two pieces are orthogonal and sum to $\mathbf{a}$.

Exercise 2.5 (★★core — a perpendicular direction). Find a vector orthogonal to both $\mathbf{a} = (1, 2, 1)$ and $\mathbf{b} = (2, -1, 3)$, and then a unit vector in that direction.

Exercise 2.6 (★★core — area of a triangle in space). Find the area of the triangle with vertices $A(1, 1, 0)$, $B(2, 3, 1)$ and $C(0, 2, 2)$.

Exercise 2.7 (►★★core, physics — torque and work). A force $\mathbf{F} = (2, 3, 1)$ (newtons) acts at the point with position vector $\mathbf{r} = (1, 0, 2)$ (metres) measured from a pivot.

- Find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about the pivot.
- The point of application then undergoes the displacement $\mathbf{d} = (0, 4, 0)$ (metres). Find the work $W = \mathbf{F} \cdot \mathbf{d}$ done by the force.

Exercise 2.8 (►★★core — a line, and points on it). Let ℓ be the line through $A(1, 0, 2)$ and $B(3, 4, 6)$.

- Give a parametric equation of ℓ , using the simplest integer direction.
- Does $C(0, -2, 0)$ lie on ℓ ?

- (c) Does $D(2, 1, 4)$ lie on ℓ ?

Exercise 2.9 (► ★★ core — *the plane through three points (anchor)*). Let $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$.

- (a) Find the equation of the plane through A, B, C in point-normal and in Cartesian form, and verify that all three points satisfy it.
- (b) Write the same plane in parametric form $\mathbf{r} = \overrightarrow{OA} + s\overrightarrow{AB} + t\overrightarrow{AC}$. Evaluate the point at $s = t = 1$ and check that it satisfies your Cartesian equation.

Exercise 2.10 (► ★★ core — *where a line meets a plane*). Use the plane $3x + 2y + z = 6$ of Exercise 2.9.

- (a) Where does the line $\mathbf{r} = t(1, 1, 1)$ meet the plane?
- (b) Does the line $\mathbf{r} = t(1, -1, -1)$ meet the plane? Explain your answer using the dot product of the line's direction with the plane's normal.

3 Further practice (home)

Exercise 3.1 (★★core, data science — ranking by cosine similarity). A query is the vector $\mathbf{q} = (1, 1, 1)$. Three documents are $\mathbf{d}_1 = (2, 2, 2)$, $\mathbf{d}_2 = (1, 1, 0)$, $\mathbf{d}_3 = (0, 0, 5)$. Rank them by cosine similarity to $\mathbf{q}$, and comment on $\mathbf{d}_1$.

Exercise 3.2 (★★core — unit normal and equation of a plane). Let $P(1, 0, 1)$, $Q(0, 2, 3)$ and $R(2, 0, 0)$.

- Find a unit normal to the plane through P , Q , R .
- Find the Cartesian equation of the plane.

Exercise 3.3 (★★core — angle between two planes). Find the acute angle between the planes $x + y + z = 1$ and $x + z = 0$.

Exercise 3.4 (★★core — foot and distance from a point to a plane). Let $P(3, 1, 2)$ and the plane $2x - 2y + z = 4$.

- Find the foot F of the perpendicular from P to the plane.
- Hence find the distance from P to the plane.

Exercise 3.5 (★★core, physics — work done sliding down a ramp). A crate of mass 10 kg slides down a ramp, undergoing the displacement $\mathbf{s} = (6, -8)$ (metres), with the $+y$ axis pointing up. Taking $g = 10 \text{ m/s}^2$, the weight is the force $\mathbf{F} = (0, -100)$ (newtons).

- Find the work $W = \mathbf{F} \cdot \mathbf{s}$ done by gravity.
- Find the scalar component $\mathbf{F} \cdot \hat{\mathbf{s}}$ of the weight in the direction of motion.

Exercise 3.6 (★★core — expansion identities, the parallelogram law and the rhombus). Using only $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$, the distributive rule $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$, the scalar rule $(\lambda \mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v})$ and commutativity $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$:

- show that $\|\mathbf{u} \pm \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2$;
- deduce the parallelogram law $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$, and say what it means for the diagonals and sides of a parallelogram;
- a rhombus has sides $\mathbf{u}, \mathbf{v}$ with $\|\mathbf{u}\| = \|\mathbf{v}\|$ and diagonals $\mathbf{u} \pm \mathbf{v}$; prove that its diagonals are perpendicular.

Exercise 3.7 (★★★challenge — coplanarity of four points). Let $P(0, 0, 0)$, $Q(1, 1, 0)$, $R(1, 0, 1)$.

- Decide whether P, Q, R and $S(0, 1, 1)$ are coplanar. If not, give the volume of the parallelepiped on $\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$.
- Decide whether P, Q, R and $T(2, 1, 1)$ are coplanar.

Exercise 3.8 (★★★challenge — foot of the perpendicular to a line). Let ℓ be the line through the origin in the direction $\mathbf{d} = (1, 2)$, and let $P(5, 0)$. Find the foot F of the perpendicular from P to ℓ , and the distance from P to ℓ .